

Assignment 5: SQL Advanced Queries

Name: Your Name | Course: SQL | Submission: Assignment 5

Overview

This assignment builds on SQL skills by creating a Products table and writing targeted queries covering filtering, pattern matching, and range conditions to extract insights from product data.

1. Create and Populate Table

SQL Query:

```
CREATE TABLE Products (  
    ProductID INT PRIMARY KEY,  
    ProductName VARCHAR(50),  
    Category VARCHAR(50),  
    Price DECIMAL(10,2),  
    Stock INT  
);  
  
INSERT INTO Products (ProductID, ProductName, Category, Price, Stock) VALUES  
(1, 'Laptop', 'Electronics', 800.00, 50),  
(2, 'Smartphone', 'Electronics', 600.00, 30),  
(3, 'Desk Chair', 'Furniture', 120.00, 100),  
(4, 'Table', 'Furniture', 200.00, 20),  
(5, 'Notebook', 'Stationery', 5.00, 500),  
(6, 'Pen', 'Stationery', 2.00, 1000);  
  
SELECT * FROM Products;
```

Output (screenshot_1):

Query Result – Table Created & Populated				
ProductID	ProductName	Category	Price	Stock
1	Laptop	Electronics	800	50
2	Smartphone	Electronics	600	30
3	Desk Chair	Furniture	120	100
4	Table	Furniture	200	20
5	Notebook	Stationery	5	500
6	Pen	Stationery	2	1000

2. SQL Query Tasks

a) Electronics & Furniture

Retrieve products in categories Electronics or Furniture.

SQL Query:

```
SELECT * FROM Products
WHERE Category IN ('Electronics','Furniture');
```

Output (screenshot_2a):

2a) Electronics & Furniture				
ProductID	ProductName	Category	Price	Stock
1	Laptop	Electronics	800	50
2	Smartphone	Electronics	600	30
3	Desk Chair	Furniture	120	100
4	Table	Furniture	200	20

b) Price Range

Find products with price between 100 and 800.

SQL Query:

```
SELECT * FROM Products
WHERE Price BETWEEN 100 AND 800;
```

Output (screenshot_2b):

2b) Price Range 100–800				
ProductID	ProductName	Category	Price	Stock
1	Laptop	Electronics	800	50
2	Smartphone	Electronics	600	30
3	Desk Chair	Furniture	120	100
4	Table	Furniture	200	20

c) Stock Range

List products whose stock is between 50 and 500.

SQL Query:

```
SELECT * FROM Products
WHERE Stock BETWEEN 50 AND 500;
```

Output (screenshot_2c):

2c) Stock Range 50–500

ProductID	ProductName	Category	Price	Stock
1	Laptop	Electronics	800	50
3	Desk Chair	Furniture	120	100
5	Notebook	Stationery	5	500

d) Name Contains 'Pen'

Select products with 'Pen' anywhere in the name.

SQL Query:

```
SELECT * FROM Products
WHERE ProductName LIKE '%Pen%';
```

Output (screenshot_2d):

2d) Name Contains 'Pen'

ProductID	ProductName	Category	Price	Stock
6	Pen	Stationery	2	1000

e) Name Starts With 'S'

Retrieve products whose names start with 'S'.

SQL Query:

```
SELECT * FROM Products
WHERE ProductName LIKE 'S%';
```

Output (screenshot_2e):

2e) Name Starts With 'S'

ProductID	ProductName	Category	Price	Stock
2	Smartphone	Electronics	600	30

f) Category & Price

Products in Stationery or Furniture and price 100-300.

SQL Query:

```
SELECT * FROM Products
WHERE Category IN ('Stationery','Furniture')
AND Price BETWEEN 100 AND 300;
```

Output (screenshot_2f):

2f) Category & Price (Stationery/Furniture, 100–300)

ProductID	ProductName	Category	Price	Stock
3	Desk Chair	Furniture	120	100
4	Table	Furniture	200	20

g) Low-Price 'o'

Products priced 1-10 and name contains 'o'.

SQL Query:

```
SELECT * FROM Products
WHERE Price BETWEEN 1 AND 10
AND ProductName LIKE '%o%';
```

Output (screenshot_2g):

2g) Low-Price 'o' (1–10, name contains 'o')

ProductID	ProductName	Category	Price	Stock
5	Notebook	Stationery	5	500